

HITIKA HEMNANI

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EDUCATION

University of Southern California, Los Angeles, California

Aug 2024 - May 2026

Master of Science in Computer Science

(GPA 3.8/4)

Relevant coursework: Analysis of Algorithms, Database Systems, Information Retrieval and Web Search Engines

Vivekanand Education Society's Institute of Technology, Mumbai, India

Jun 2018 - May 2022

Bachelor of Engineering in Computer Science

(CGPA 9.68/10)

Relevant coursework: Data Structures, OOP, and Databases

TECHNICAL SKILLS

- Programming: Java, Python, JavaScript (Node.js), TypeScript
- Web & APIs: React.js, Django, REST APIs, Angular
- Databases: SQL, NoSQL, MongoDB, DynamoDB, Snowflake
- Systems & Cloud: Distributed Systems, AWS, Docker, Kubernetes, CI/CD
- Tools: Linux, Debugging, CI/CD

PROFESSIONAL EXPERIENCE

USC Auxiliary Services, Los Angeles, CA

Sep 2025 - May 2026

Software Developer – Part-time

- Designing and building internal software to automate operational workflows, including full-stack development, API integrations, and data pipelines that replace manual processes as part of a broader push to modernize the department's tech infrastructure.

University of Southern California, Los Angeles, CA

Feb 2025 - Dec 2025

Research Assistant – Software Engineer

Mental Health Project

- Designed and optimized a modular EEG data processing and inference pipeline, enabling near real-time deployment and improving computational efficiency by 22%.
- Built automated workflows for large-scale data preprocessing and configuration, reducing manual setup time by 30%.
- Developed a React-based dashboard for visualization and system monitoring.

LTI Mindtree, Mumbai, India

Jun 2022 - Mar 2024

Software Engineer – SAP Business Unit

- Developed and debugged backend logic and integration scripts, enhancing system scalability across 20+ global regions.
- Partnered with global enterprise clients to deliver scalable backend solutions and automate HR workflows, improving reliability and performance by 30%.
- Collaborated in Agile sprints, contributing to sprint planning, code reviews, and deployment cycles

Trivia Software, Thane, India

May 2021 - Jun 2021

Software Developer Intern

- Developed and deployed a responsive full-stack web application using Django and React/TypeScript, reducing page load time by 40%.
- Integrated REST APIs with frontend state management for dynamic, real-time data updates.
- Implemented PyTest and Jest frameworks for complete functional testing in CI/CD pipelines.

ACADEMIC PROJECTS AND PAPERS

Down Syndrome Detector | React, Django, Python, SQL - [Publication](#)

- Built a health-assistant web app with a Django backend and secure REST APIs for real-time predictions.
- Designed accessible user flows and dashboards that improved navigation and engagement for non-technical users.
- Documented and published the system for its contribution to improving healthcare accessibility.

Drone Search and Rescue | Python, AirSim, Unreal Engine

- Implemented voxel-based 3D exploration and LiDAR-based obstacle avoidance for autonomous drone missions.
- Simulated scalable algorithms for victim detection and prioritized search paths.

Transfer Learning for Waste Image Classification | Python, Keras, TensorFlow, OpenCV - [GitHub](#)

- Implemented and compared VGG16, ResNet, and EfficientNet; achieved AUC 0.96 with VGG16.
- Performed error analysis and hypothesis testing to evaluate performance gains.

University Canteen Management System | React, Django, Python, JavaScript, SQL - [Publication](#)

- Built a full-stack canteen management web app with a React frontend and Django backend to enable students to place real-time food orders and receive personalized dish recommendations.
- Optimized UI rendering and API communication paths, reducing checkout time by 35%.
- Designed data-driven features that adapt to users' order history to improve overall user experience.

EXTRACURRICULAR ACTIVITIES

- Published two peer-reviewed papers in Springer and IJCRT on AI in healthcare and smart systems.
- Completed JPMorgan's virtual software engineering program focused on data visualization.
- Awarded Top 5% of Class, V.E.S.I.T, Mumbai.